

Assignment 1 R

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1 The Welch Two-Sample t-test

To perform the statistical analysis, the function `Welch.t.test` was implemented. The implementation follows a three-step process: input validation, iterative computation, and object construction.

Input Validation

The function begins by verifying the integrity of the inputs. The input consists of a tibble or data frame where the columns represents the different variables, a group vector of containing only A or B which indicates which group each observation belongs to, a vector consisting of characters which tells the names of the variables, significance level which is set default to 0.05, a logical value which indicates if bonferroni correction should be applied or not. The function ensures that the provided data is a data frame or tibble, that the group vector exclusively contains "A" and "B", and that the length of the names vector corresponds to the number of columns in the dataset.

Computation

The core analysis is performed by implementing a for-loop to iterate through each variable (column) defined in the names argument. For each iteration, the column data is first split into two distinct samples based on the values "A" or "B" in the group vector. As required, the test statistics are calculated manually. For each variable, the sample size (n_A, n_B), the sample mean ($\bar{X}, \bar{Y}$), and the sample variance (s_X^2, s_Y^2) are computed through:

$$\bar{X} = \frac{1}{n_A} \sum_{i=1}^{n_A} X_i, \quad \bar{Y} = \frac{1}{n_B} \sum_{j=1}^{n_B} Y_j$$

$$s_X^2 = \frac{1}{n_A - 1} \sum_{i=1}^{n_A} (X_i - \bar{X})^2, \quad s_Y^2 = \frac{1}{n_B - 1} \sum_{j=1}^{n_B} (Y_j - \bar{Y})^2$$

Using these components, the Welch t-statistic and the degrees of freedom (ν) are computed via.

$$t = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_X^2}{n_A} + \frac{s_Y^2}{n_B}}}$$

$$\nu = \frac{\left(\frac{s_X^2}{n_A} + \frac{s_Y^2}{n_B}\right)^2}{\frac{(s_X^2/n_A)^2}{n_A - 1} + \frac{(s_Y^2/n_B)^2}{n_B - 1}}$$

Finally, the p-value is obtained with the function `pt()`. If requested, a Bonferroni correction is applied before the function returns the output. The output consists of the computed values, the name vector, the significance level, the input tibble or data frame, and the group vector. This structure ensures that the print and plot methods have all the necessary information to format the output. The last step is to assign the output to the class `Welch_t_test`. This classification is critical because it enables the use of other functions on the results.

1.1 Print

The first method `print.Welch_t_test` makes sure the output from the first function is tidier. The print function prints out a table with rows giving 1. Variable names, 2. P-values and 3. t-statistics in that order. The function uses three decimals for the p-value and four decimals for the t-value. The number of decimals is chosen not to make it overwhelming for the user. The print function also prints the significance level to make it easier to define significance for the user. The purpose of the first method is to make it easier to read the output from the first function.

1.2 Plot

The plot function `plot.Welch_t_test` constructs a scatter plot from the library `ggplot`. Why `ggplot` is used is because it has more options for plots. The scatterplot consists of the observations of each variable. The plot includes label names where the x-axis represents the variables and the y-axis represents the variable values. The plot also separates the groups and color-codes them, with the different colors explained in a separate legend. The purpose of the second method is to obtain a clean and tidy visualization of the data for the user.

2 2016 U.S Presidential Election

2.1 Goal of the report

The goal of the report is to investigate in which type of counties Trump was more popular. To do this, 3 datasets were used.

2.1.1 General results

This dataset contains the presidential results of 2012 and 2016 for each country in the US.

2.1.2 County facts

This dataset contains demographic data from each country, taken from the US Census Bureau.

2.1.3 Crime Data

This dataset contains reported crimes per country taken from the Inter-university Consortium for Political and Social Research (ICPSR), Institute for Social Research, University of Michigan.

2.2 Merging the datasets

First, observations with wrong or incomplete FIPS codes were removed. In crime data, the column "FIPSC" had observations with FIPS codes 999 and 777, which meant it was not possible to create a complete FIPS code since these did not seem to correspond to any real county code at all. This resulted

in 35 removed observations from the dataset. After that, observations with FIPS 00000 in county facts were removed, resulting in one more removed observation. In the facts variable, there were also variables corresponding to a state summary instead of a specific county. These were also removed, resulting in 51 removed variables. In summary, the initial data cleaning resulted in 87 removed observations.

To analyze in which counties Trump was more popular, a left join with the main variable election results was chosen for the analysis by merging the dataset by FIPS code. After ensuring all datasets had the same FIPS format and after initial data cleaning, all datasets were merged. When checking which variables were lost after the merge, 6 unique variables were found.

state_abbr	county_name	FIPS
AK	Alaska	02068
AK	Alaska	02105
AK	Alaska	02195
AK	Alaska	02198
AK	Alaska	02230
AK	Alaska	02275

Figure 1: Lost observations after left join

When checking these, all variables that were lost belonged to the state of Alaska. These were also exact duplicates except for the FIPS code, meaning something wrong must have occurred with these observations.

When checking the final merged dataframe, we check for more duplicates to avoid errors in the analysis. Duplicates in one column, for example, GOP votes 2016 could be possible, but in several columns at the same time, like total votes, democrat votes, and "difference in votes" is unreasonable. When checking this, 29 duplicates were found. These also corresponded to Alaska with the same values in each column. Therefore, all of these were removed. After this, when checking for duplicates again, 0 were obtained.

In total, the initial data cleaning, duplicate removal, and missing FIPS codes after merge resulted in 122 removed observations in total. The final dataset consisted of 3112 observations.

2.3 Percentage of GOP votes

The percentage of GOP votes is the main variable that will be studied to explore in which type of counties Trump was more popular. This variable measures the percentage of GOP (Republican Party) votes per county.

	Percentage of GOP votes (%)
Min	4
1st Qu.	55
Median	67
Mean	64
3rd Qu.	75
Max.	95

Looking closer at the variable, the mean percentage of GOP votes was around 64%. The range of the data is wide, with a minimum value of around 4% and a maximum of around 95%. The difference between the first and third quantiles is, however, only 20 percentage points, illustrating that much of the variability comes from the highest and lowest 25% of the data.

2.4 Exploratory Data Analysis

To begin to explore which variables might be of interest and in which type of counties Trump was more popular, an exploratory correlation analysis was performed between the variable "percentage of GOP votes" (being the party for Trump) and the other numerical values in the data frame. Doing this resulted in the following highest correlation values, when ignoring the variables that serve as alternative measures to the percentage of GOP votes and the variable itself :

Variables	Positive correlation
White alone, not Hispanic or Latino, percent, 2014	0,53
White alone, percent, 2014	0,52
Variables	Negative correlation
Housing units in multi-unit structures, percent, 2009-2013	0,6
Bachelor's degree or higher, percent of persons age 25+, 2009-2013	0,49
Robbery rate	0,48

Since white alone and white alone not Hispanic or Latino were very similar in correlation, and since the white alone without specifying further might be harder to interpret, white alone not Hispanic or Latino was chosen. Therefore, the following variables were investigated further:

Variables
White alone, not Hispanic or Latino, percent, 2014
Bachelor's degree or higher, percent of persons age 25+, 2009-2013
Housing units in multi-unit structures, percent, 2009-2013
Robbery rate

2.5 Education

	Percentage with bachelor's degree or higher, percent of persons age 25+, 2009-2013 (%)
Min	3,2
1st Qu.	13,7
Median	17,5
Mean	19,7
3rd Qu.	23,4
Max.	74,4

Looking closer at the education variable, the mean with a bachelor's degree or higher is around 19 percent. However, the summary indicates a large variation between counties, with the smallest one only having 3 percent and the biggest one having around 74 percent, which is substantially bigger than the median.

Education Groups	Number of counties	Mean in percentage of GOP votes %
Low Education (0-15%)	1066	68,2
Moderate Education (15-25)	1398	66,4
High Education (25-40%)	525	52,6
Very High Education (40-100%)	123	38,3

Based on the mean of the variable, the data was divided into 4 groups based on percentage with a bachelor's degree or higher. The lower and moderate education groups were placed below and around the mean, while the high and very high education groups were placed over the mean.

Looking closer at the difference between the groups of education, the mean decreases quite dramatically towards the high and very high education groups, while the difference between low and moderate is smaller. This indicates that Trump was more popular in counties with a smaller percentage of people with a bachelor's degree or higher.

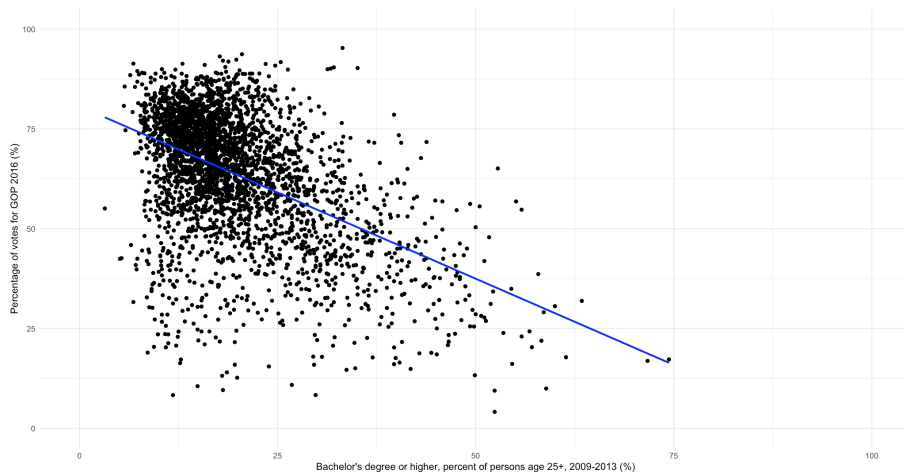


Figure 2: Connection between bachelors degree or higher and percentage of GOP votes

To study the connection further, a scatterplot was drawn between the percentage of GOP votes in 2016 and the percentage of persons in each county with a bachelor's degree or higher. The majority of dots are located around the left corner, indicating a lower percentage with a bachelor's degree and a higher percentage of Trump votes. It looks like Trump was more popular in counties with a lower percentage of education and less popular in counties with higher education. However, here the spread across the line is quite big, indicating that other forces impacting voting have to be studied.

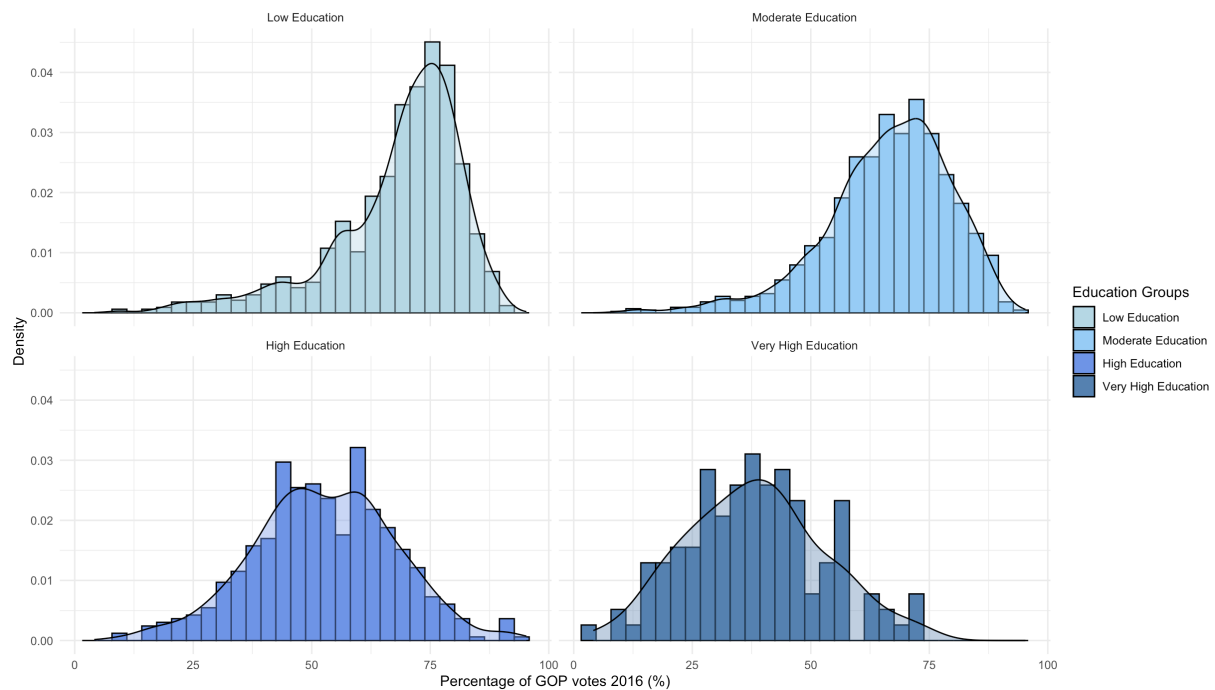


Figure 3: Voting variability between different education groups

To study the variability between the groups, a faceted histogram was created. Since the groups are of different size, the density is shown in the y axis instead of the number of counties. Here, it can be seen that the density is noticeably higher at higher percentages of GOP votes across the lower education groups, while the variability is much higher across the higher education category, where the higher percentages in GOP votes also have lower density. The plot therefore seems to indicate that Trump was more popular across counties in the lower education groups, while the voting pattern was more divided across the higher education groups.

2.6 White alone, not Hispanic or Latino, percent, 2014

	White alone, not Hispanic or Latino, percent, 2014 (%)
Min	3,1
1st Qu.	65,7
Median	84,7
Mean	77,4
3rd Qu.	93,1
Max.	98,6

Looking closer at the summary statistics for the variable "White alone, not Hispanic or Latino, percent, 2014" shows a large skewness in the demographic. A clear majority of the counties have a large percentage of white people, with a mean percentage of around 77,4%. However, the minimum value starts at 3% and ends at around 99% while the median is around 84,7%, illustrating a spread in this variable too.

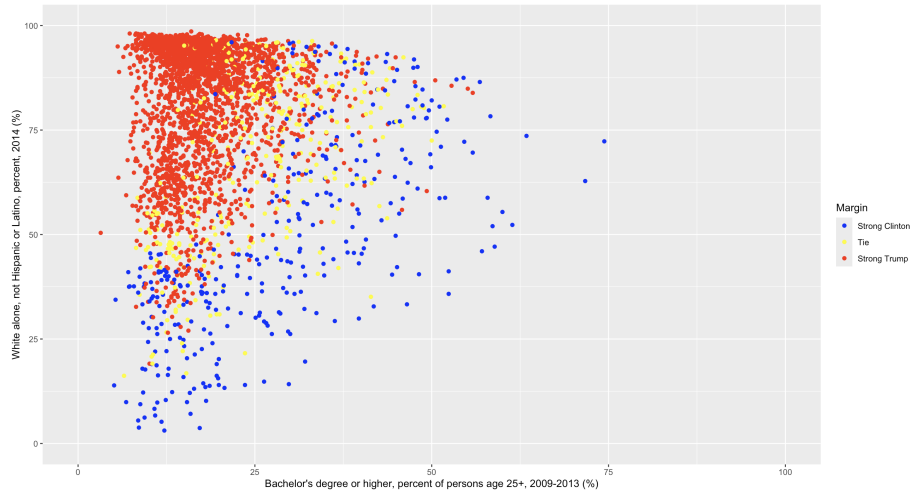


Figure 4: Relationship between education and bachelor's degree, colored by win margin

By dividing the variable that measured win margin for Trump or Clinton where more than 10 percent or more win was considered a strong Trump or Clinton win, while a smaller margin was considered a tie, a third aesthetic was plotted with the percentage of white people (not Hispanic or Latino) and percentage with bachelor's degree or higher, together with color illustrating strong win or tie.

This shows what was shown earlier that Trump seems to be more popular across counties with a lower percentage with a bachelor's degree or higher. However, here it can also be seen that Trump also appeared to be more popular across counties with a higher percentage of white people, while the strong Clinton dots are more spread out across the right, with more dots across the higher percentage with a bachelor's degree and a lower percentage of white people.

Since a clear majority of counties had a high percentage of white people (not Hispanic or Latino), the percentage of white people variable was divided into four nearly equally large groups (quantiles) to further investigate the connection between Trump votes and the percentage of white people in each county.

Race Homogeneity	Mean in percentage of GOP votes
Most diverse (5,9-80,7%)	53
Modereratly Diverse (80,7-92%)	62,9
Majority White (92-96%)	68
Near all White (96-99,3%)	71

This shows what was already shown earlier, that the data is extremely left-skewed and that counties with a smaller percentage of white people are rarer. The most diverse group and the moderately diverse group get split into 5,9-92% white. Still, after the split, a notable difference is shown in the mean between the most diverse counties and the counties that are all near white, with the mean in GOP votes steadily increasing with each group.

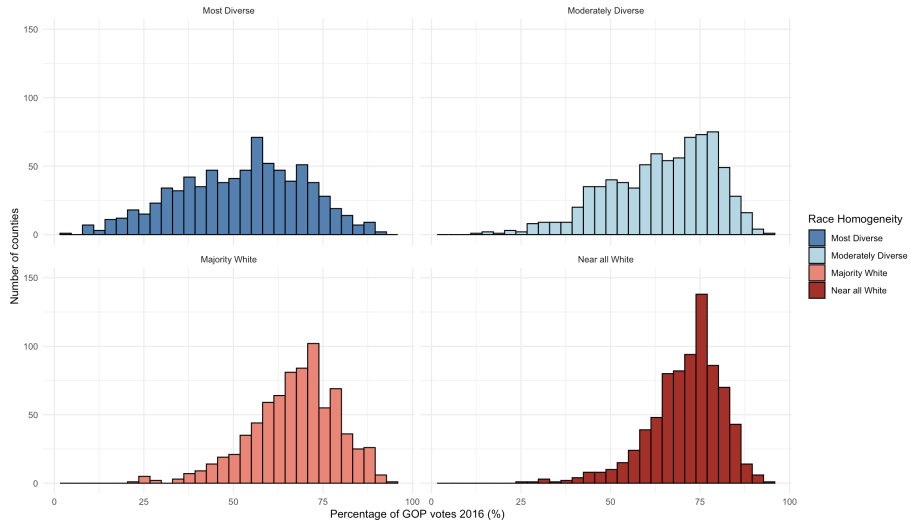


Figure 5: Difference in voting between race homogeneity groups

This can be further illustrated if the four groups are illustrated in four histograms. Here, it can also be seen that the distribution of GOP vote percentage becomes increasingly more skewed to the left when the percentage of white people increases, while the variability is the largest in the groups with a lower percentage of white people in the most and moderately diverse groups.

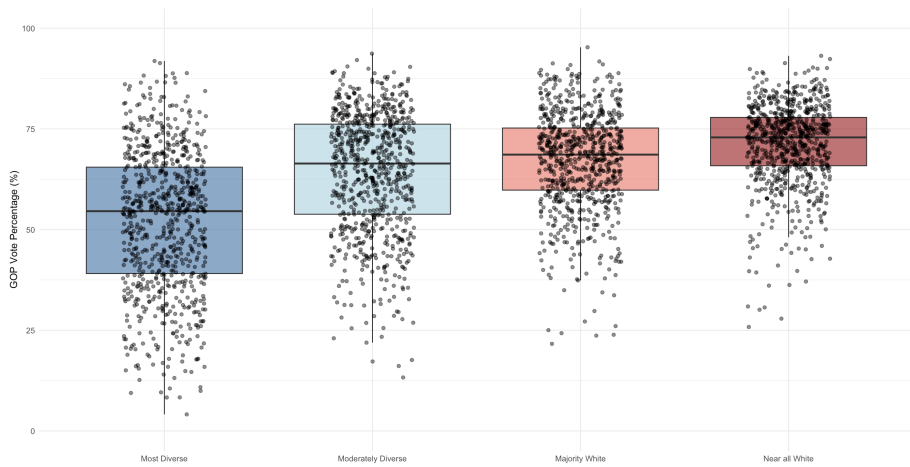


Figure 6: Voting variability between different race homogeneity groups

Illustrating the four groups in a jittered boxplot illustrates a similar pattern with a more concentrated and higher median GOP votes for the majority white and nearly all white counties. The variability also decreases in each group, with the lowest variability across the near all white group. However, the overall range is still large across all groups, with counties voting both in high and low percentages for Trump, indicating that even if the data becomes more concentrated, many other factors need to be evaluated.

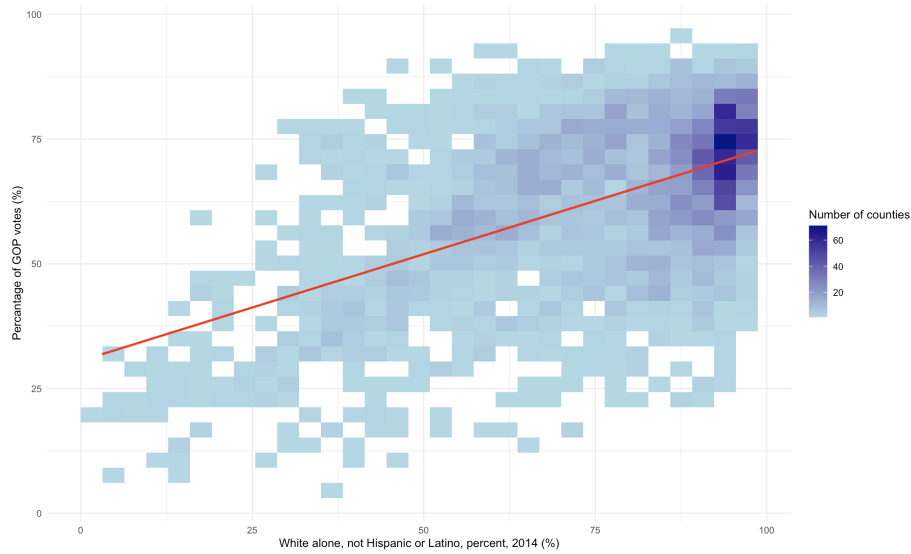


Figure 7: Relationship between Racial decomposition and percentage of GOP votes

As mentioned earlier, a majority of the counties had a majority of white percentage (not Hispanic or Latino), which can also be seen in the following heatmap. Looking at the heatmap, a majority of the counties with a majority voting for Trump are located near a higher percentage of white people, while a lower percentage of white people are in fewer counties, and also a lower percentage of GOP votes. However, there is a large variability across the line, and it is hard to see a clear connection here.

To summarize, it seems that Trump was increasingly more popular in counties with a higher percentage of white people, while the more diverse counties had a bigger variability in how they voted, with a lower mean in Trump votes. However, the plots also show variability across all groups, confirming that many other factors affect the voting.

2.7 Housing units in multi-unit structures, percent, 2009-2013

The housing units in multi-unit structures variable could be a good measurement of urbanization and how densely populated an area is, which might be why this variable had a high correlation. This might indicate that Trump was more popular in less urbanized and less densely populated areas.

Housing units in multi-unit structures, percent, 2009-2013	
Min	0
1st Qu.	6,2
Median	9,7
Mean	12,3
3rd Qu.	15,8
Max.	98,5

Looking at the summary statistics for the variable, the mean is low, around 12 percent, while the maximum is around 98 percent, indicating that the extremely high values pull up the mean. Since the data seems skewed with a majority of the counties having a mean of percentage of housing units in multi-unit

structures near 0, we divide the data into 5 near equally large groups (quantiles) and compare the means of the groups. This results in the following table:

Multi-Unit Housing Groups	Mean in percentage of GOP votes
Very Low Multi-unit Housing (0 - 5,5%)	72,11
Low Multi-unit Housing (5,5 - 8,2%)	70,47
Moderate Multi-unit Housing (8,2 - 11,6%)	66,1
High Multi-unit Housing (11,6 - 17,78%)	61,56
Very High Multi-unit Housing (17,78 - 98,5%)	47,75

The limits of the groups confirm that the data is very skewed to the right, with a majority of the counties having a smaller percentage of multi-unit housing. Still, when divided into five groups, the mean in GOP votes seems to slowly decrease, with a smaller percentage of GOP votes across the higher multi-unit housing groups and a quite large difference between the highest and lowest group of around 25 percentage points. This variable could be a good indicator of high vs low urbanization, with a higher percentage of multi-unit housing in bigger cities and a lower percentage of multi-unit housing in smaller cities.

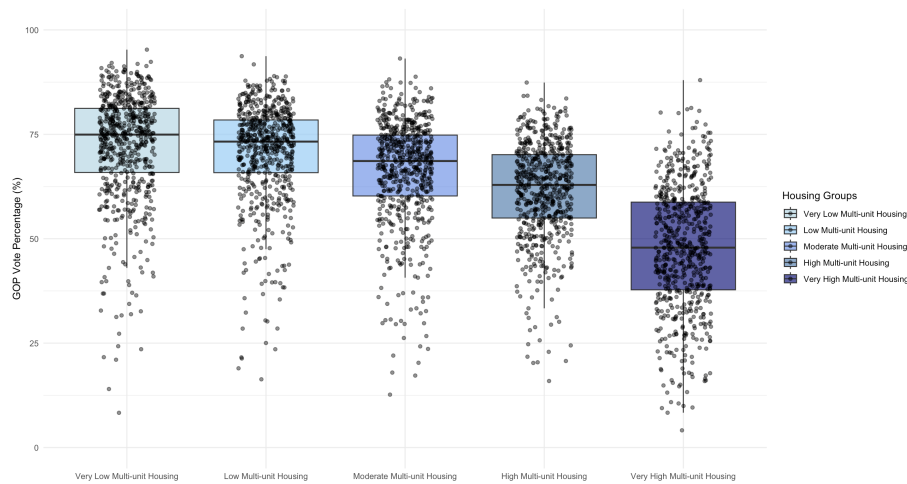


Figure 8: Difference in variability in percentage of GOP votes between different housing groups

Looking at the divided boxplot for the groups shows that the percentage of GOP votes had the highest median across the lower percentage multi-unit housings, while the median was lower across the higher percentage multi-unit housing. This might indicate that Trump was more popular in smaller, less densely populated counties that tended to have a lower percentage of multi-unit housing compared to more densely populated areas with a higher percentage of multi-unit housing. The variability is also the largest across the very high percentage of multi-unit structures group, while the other groups have quite similar variability.

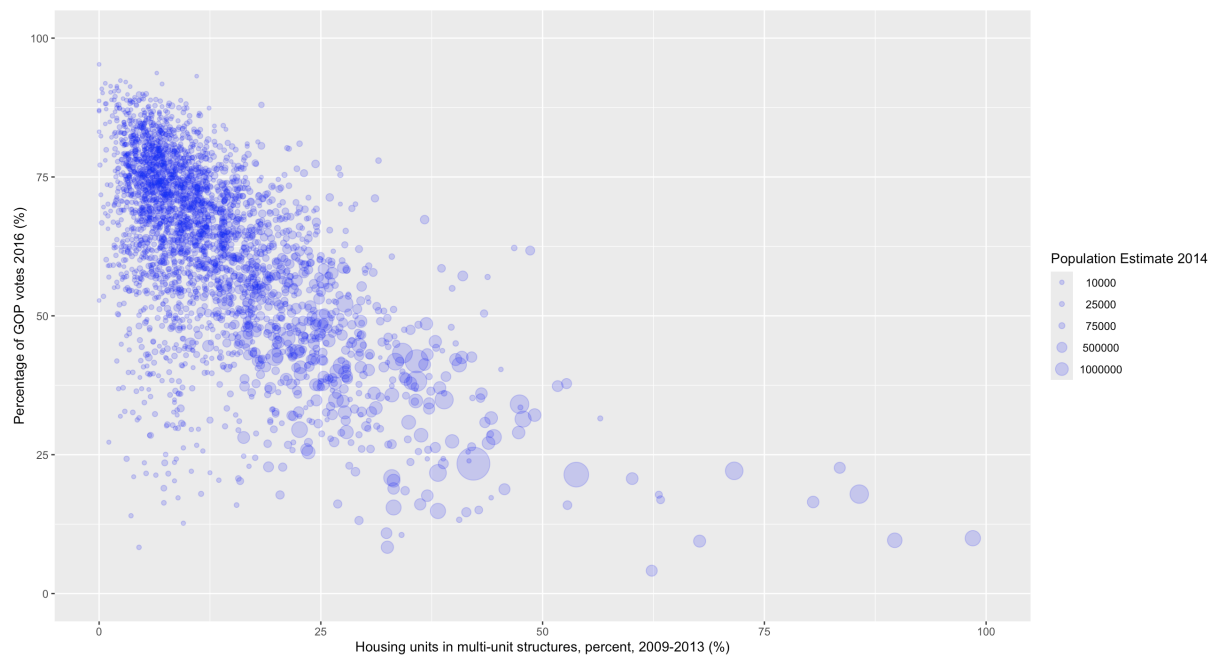


Figure 9: Relationship between GOP Vote Share and Multi-Unit Housing Percentage, sized by Population

To evaluate if the percentage of multi-unit housing and population size could be a good measure of urbanization, and if this seems to affect voting, a bubble plot was created with bubble size corresponding to the population estimate of 2014, together with the percentage of housing units and the percentage of GOP votes.

The majority of counties where Trump was more popular were in the left corner with a smaller percentage of multi-unit structures and smaller bubbles. The bigger bubbles tended to be more spread out across the right and downwards, with a bigger percentage of multi-unit structures and a lower percentage in Trump votes. This indicates that multi-unit structures and population size might be good indicators of the level of urbanization, and that more urbanized counties with bigger population and higher percentage of multi-unit structures tend to favor Clinton, while less urbanized counties tend to favor Trump.

2.8 Robbery rate

Robbery rate per 100 000	
Min	0
1st Qu.	0
Median	12,99
Mean	29,760
3rd Qu.	36,4400
Max.	595,86

Finally, looking at the last variable robbery rate shows that the mean in robbery rate is low, around 30 per 100,000 inhabitants. This variable had a negative correlation, which is interesting since Trump

campaigned a lot around law and order.

	Percentage of GOP votes (%)
Higher Robbery Rate (Upper limit)	58,3
Lower Robbery Rate (Lower limit)	68,9

Dividing the robbery rate in two groups, one above and one below the median, which was around 13 per 100,000 habitants shows that the mean of GOP votes was higher for the lower robbery rate compared to the higher robbery rate.

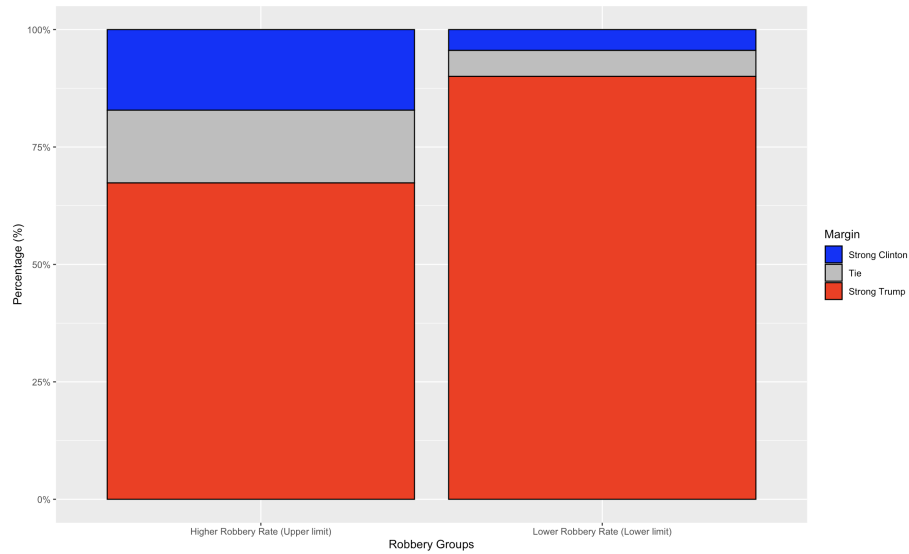


Figure 10: Division of votes between Low and High Robbery rate

The following stacked barplot was created with win margin to compare the two robbery rate groups. The strong Trump counties had a higher percentage across the low robbery rate compared to the high robbery rate. Could this just be that the robbery rate was higher across more densely populated areas where Clinton was more popular?

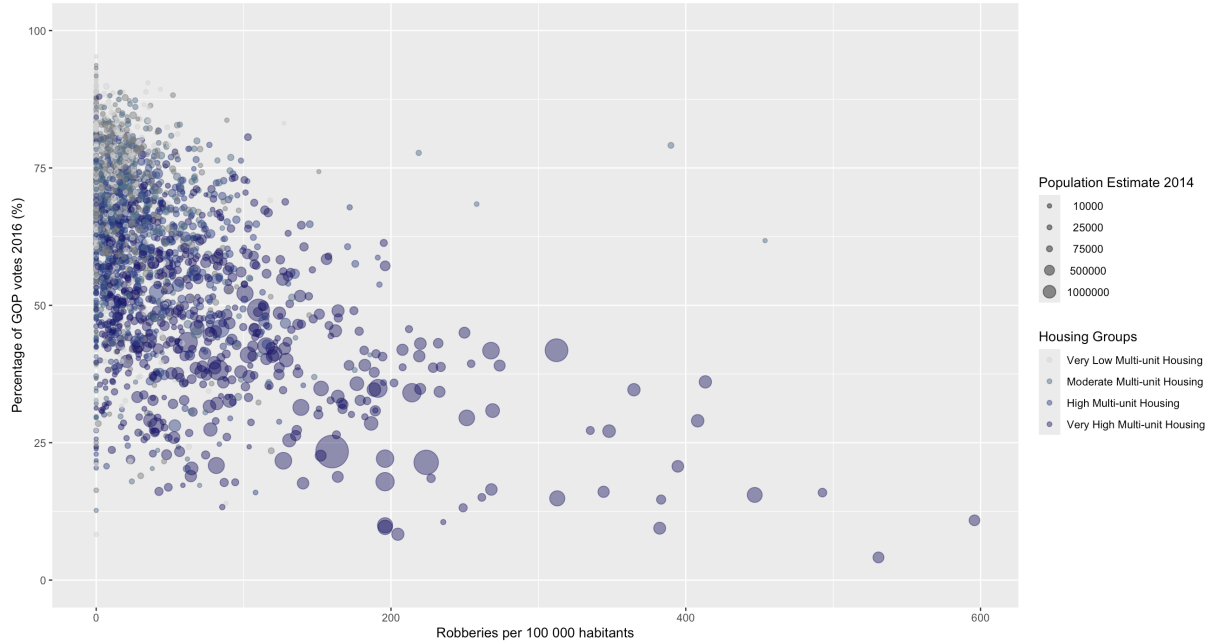


Figure 11: Relationship between GOP Vote Share and Robberies per 100 000, sized by Population and colored by Housing Group

To further evaluate this, a bubble plot was created with housing groups, population size, robberies, and percentage of GOP votes.

Here, the connection can be seen more clearly. Nearly all small points with lighter colors indicating less densely populated areas are near the low robbery rate and higher percentage of GOP votes, while the bigger points in darker colors, illustrating higher percentages of multi-unit housing and bigger population, tend to be more spread out across higher intensity of robbery rates and lower percentage Trump votes.

This strengthens the theory that the negative correlation between Trump votes and robbery rate is actually driven by multi-unit housing and population, indicating urbanization, rather than that robberies are the dominant factor in how people voted. Trump appears to be more popular in less urbanized counties. The less urbanized counties tend to have smaller populations and a lower percentage of multi-unit housing, which most likely also contributes to fewer robberies because of less densely populated areas, therefore probably creating the correlation seen.

2.9 Conclusion

In conclusion, Trump appears to be more popular in less urbanized counties. The variables containing the percentage of multi-unit structures showed a strong connection with population estimate, indicating that these variables were a good measurement for urbanization. When the percentage of multi-unit housing was divided into four 5 near equally large groups (quantiles), Trump appeared more popular across less urbanized counties with a lower percentage of multi-unit housing and a smaller population.

Interesting patterns were also seen between Trump votes and education, with a more concentrated higher percentage of votes across the lower education groups compared to the higher education groups, where the variability in Trump votes was higher.

A similar pattern was also seen when studying a possible connection between the percentage of white people (not Hispanic or Latino) and Trump votes. Even though a majority of counties had a higher percentage of white people, when dividing into four nearly equal groups (quantiles), the most diverse groups had a bigger variability in how they voted, compared to the groups with the highest percentage of white people, which had a more concentrated higher Trump voting percentage. However, even though the voting seemed more concentrated across a higher percentage for Trump, there was still variability across all groups. The same thing could be said for education groups, too, with a variability in voting patterns across all groups here, too.